

A Practical Guide to Image captioning

TOPIC -- IMAGE CAPTIONING

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Creative writing and computational humor Creative language generation by NLG has been hypothesized since the field's origins. A recent pioneer in the area is Phillip Parker, who has developed an arsenal of algorithms capable of automatically generating textbooks, crossword puzzles, poems and books on topics ranging from bookbinding to cataracts. The advent of large pretrained transformer-based language models such as GPT-3 has also enabled breakthroughs, with such models demonstrating recognizable ability for creating-writing tasks. A related area of NLG application is computational humor production. JAPE (Joke Analysis and Production Engine) is one of the earliest large, automated humor production systems that uses a hand-coded template-based approach to create punning riddles for children. HAHAAcronym creates humorous reinterpretations of any given acronym, as well as proposing new fitting acronyms given some keywords. Despite progresses, many challenges remain in producing automated creative and humorous content that rival human output. In an experiment for generating satirical headlines, outputs of their best BERT-based model were perceived as funny 9.4% of the time (while real headlines from The Onion were 38.4%) and a GPT-2 model fine-tuned on satirical headlines achieved 6.9%. It has been pointed out that two main issues with humor-generation systems are the lack of annotated data sets and the lack of formal evaluation methods, which could be applicable to other creative content generation. Some have argued relative to other applications, there has been a lack of attention to creative aspects of language production within NLG. NLG researchers stand to benefit from insights into what constitutes creative language production, as well as structural features of narrative that have the potential to improve NLG output even in data-to-text systems.

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Chatbots Another area where NLG has been widely applied is automated dialogue systems, frequently in the form of chatbots. A chatbot or chatterbot is a software application used to conduct an on-line chat conversation via text or text-to-speech, in lieu of providing direct contact with a live human agent. While natural language processing (NLP) techniques are applied in deciphering human input, NLG informs the output part of the chatbot algorithms in facilitating real-time dialogues. Early chatbot systems, including Cleverbot created by Rollo

Carpenter in 1988 and published in 1997, reply to questions by identifying how a human has responded to the same question in a conversation database using information retrieval (IR) techniques. Modern chatbot systems predominantly rely on machine learning (ML) models, such as sequence-to-sequence learning and reinforcement learning to generate natural language output. Hybrid models have also been explored. For example, the Alibaba shopping assistant first uses an IR approach to retrieve the best candidates from the knowledge base, then uses the ML-driven seq2seq model re-rank the candidate responses and generate the answer.

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Natural language generation (NLG) is a software process that produces natural language output. A widely cited survey of NLG methods describes NLG as "the subfield of artificial intelligence and computational linguistics that is concerned with the construction of computer systems that can produce understandable texts in English or other human languages from some underlying non-linguistic representation of information". While it is widely agreed that the output of any NLG process is text, there is some disagreement about whether the inputs of an NLG system need to be non-linguistic. Common applications of NLG methods include the production of various reports, for example weather and patient reports; image captions; and chatbots like ChatGPT. Automated NLG can be compared to the process humans use when they turn ideas into writing or speech. Psycholinguists prefer the term language production for this process, which can also be described in mathematical terms, or modeled in a computer for psychological research. NLG systems can also be compared to translators of artificial computer languages, such as decompilers or transpilers, which also produce human-readable code generated from an intermediate representation. Human languages tend to be considerably more complex and allow for much more ambiguity and variety of expression than programming languages, which makes NLG more challenging. NLG may be viewed as complementary to natural language understanding (NLU): whereas in natural-language understanding, the system needs to disambiguate the input sentence to produce the machine representation language, in NLG the system needs to make decisions about how to put a representation into words. The practical considerations in building NLU vs. NLG systems are not symmetrical. NLU needs to deal with ambiguous or erroneous user input, whereas the ideas the system wants to express through NLG are generally known precisely. NLG needs to choose a specific, self-consistent textual representation from many potential representations, whereas NLU generally tries to produce a single, normalized representation of the idea expressed. NLG has existed since ELIZA was developed in the mid 1960s, but the methods were first used commercially in the 1990s. NLG

techniques range from simple template-based systems like a mail merge that generates form letters, to systems that have a complex understanding of human grammar. NLG can also be accomplished by training a statistical model using machine learning, typically on a large corpus of human-written texts.

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Pollen counts are expected to remain high at level 6 over most of Scotland, and even level 7 in the south east. The only relief is in the Northern Isles and far northeast of mainland Scotland with medium levels of pollen count.

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Stages The process to generate text can be as simple as keeping a list of canned text that is copied and pasted, possibly linked with some glue text. The results may be satisfactory in simple domains such as horoscope machines or generators of personalized business letters. However, a sophisticated NLG system needs to include stages of planning and merging of information to enable the generation of text that looks natural and does not become repetitive. The typical stages of natural-language generation, as proposed by Dale and Reiter, are:

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Further reading Dale, Robert; Reiter, Ehud (2000). Building natural language generation systems. Cambridge, UK: Cambridge University Press. ISBN 978-0-521-02451-8. Evans, Roger; Piwek, Paul; Cahill, Lynne (2002). What is NLG?. INLG2002. New York, US. paper Gatt, Albert; Krahmer, Emiel (2018). "Survey of the State of the Art in Natural Language Generation: Core tasks, applications and evaluation". Journal of Artificial Intelligence Research. 61: 65–170. arXiv:1703.09902. doi:10.1613/jair.5477. S2CID 16946362. Reiter, Ehud (2018-01-16). "How do I Learn about NLG?".